

R0.72Y | full Fourier row and forced-transfer boundaries

Exact algebra, exact lift-up counterexample, and normalized forcing-rate guide



A Exact complete-row structure

The scalar invariant row closes; orthogonality alone does not close the full row.

Complete Fourier-Leray row

$$\begin{aligned} u_d &= -L u - P_j(\\ &\quad i c W u + \text{Lambda } W_x u_2 e_3) \\ \text{div}_j u &= 0 \\ L p_i &= 2 i c W_x u_2 \end{aligned}$$

pressure factor 2: CLOSED

exact

mu > 0: Orr-Sommerfeld / Squire

$$\begin{aligned} q_d &= (-L-i c W) q \\ &\quad - i c W_{xx} L^{-1} q \\ \text{eta}_d &= (-L-i c W) \text{eta} \\ &\quad + i x i \text{Lambda} W_x L^{-1} q \end{aligned}$$

q

lift-up forcing

eta

STRONG FULL-ROW A2: OPEN

Scalar invariant polarization

$$\begin{aligned} u &= g(\text{gamma}, 0, -x i) / \text{sqrt}(\mu) \\ g_d &= (\text{A_beta}^2 - \mu - i c W) g \\ \text{scalar A2 embedding: CLOSED} \end{aligned}$$

K_z=0 exact lift-up

$$u_3 = -\text{Lambda } d \exp(-x i^2 d) \times W_x(d, x) u_2(0)$$

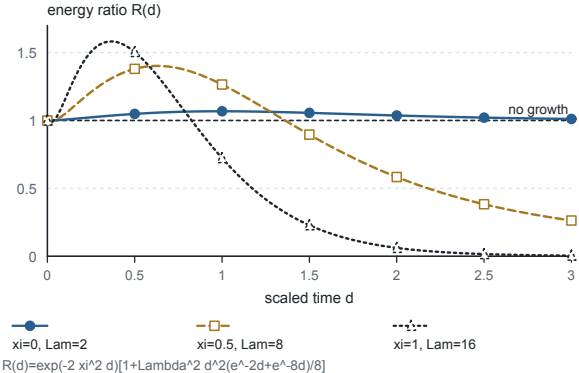
uniform strict contraction: FALSE

Velocity recovery is exact only for mu > 0; the mu=0 row is checked directly.
Direct-sum orthogonality removes row-count loss only after a uniform row bound; it does not create that bound.

B Exact zero-coupling lift-up

Energy ratio from the closed formula; line at 1 means no growth.

EXACT COUNTEREXAMPLE - NOT A STABILITY PROOF



C Scalar forced-transfer powers

Normalized operator-rate guide; constants are suppressed.

EXACT RATE GUIDE - ANALYTIC PROOF ELSEWHERE

